

Cover Letter

To: Editor-in-Chief

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Re: New submission — *Online Traveling Salesman Games: Core Fragility and the Temporal Nucleolus under Dynamic Arrivals*

Authors:

- Seyun Jeong (Korea Institute of Industrial Technology, Cheonan, Republic of Korea) — jeongsseyyun@kitech.re.kr
- Hyunchul Tae (Korea Institute of Industrial Technology, Cheonan, Republic of Korea) — *corresponding author, sage@kitech.re.kr*

Dear Editor,

We submit the enclosed manuscript for consideration as a new research article in *EJOR*. The paper extends the cooperative Traveling Salesman Game of Potters, Curiel, and Tijs (1992) to an online setting in which customers arrive over time, couples the cooperative-game stability question to the realized dispatch dynamics, and provides both analytic and computational results.

The topic sits squarely at the intersection of routing and cooperative game theory, a combination with strong *EJOR* lineage: the basic Vehicle Routing Game nucleolus is from Göthe-Lundgren, Jörnsten, and Värbrand (1996, *Mathematical Programming*); the heterogeneous-fleet extension and the time-window VRG nucleolus appeared respectively in *Transportation Science* (Engevall et al. 2004) and *Applied Mathematical Modelling* (Tae, Kim, and Park 2020); the cooperative-TSP rolling-horizon Shapley/Core variants are from Kimms and Kozeletskyi (2016a, *EJOR*; 2016b, *EURO J. Transp. Logist.*); and the most recent collaborative-routing cost-allocation paper (Lyu, Lalla-Ruiz, and Schulte, 2025, *EJOR*) provides the most direct methodological precedent — row-generation over a static coalition family, replaced here by a sequential-LP cascade over a *time-induced* family.

Core contribution

We introduce the *Online Traveling Salesman Game* $(N, c, \mathcal{F})$, a cooperative game in which the family $\mathcal{F}$ of feasibility- admissible coalitions is endogenously induced by the realized arrival-service process rather than specified exogenously as in Myerson-style graph-restricted or antimatroid-restricted games. On this family we define the *Temporal Nucleolus* as the lexicographic minimizer of excess and compute it by a sequential linear program over $\mathcal{F}$.

The paper’s theoretical contribution is a general partition-pair certificate (Theorem 14, formerly Proposition 14) for Temporal Core emptiness, with three a priori

computable specializations: the single-complement threshold r^{**} (Corollary 9, formerly Theorem 9), the balanced-complement threshold r^{***} (Corollary 10, formerly Proposition 10), and the balanced-near-complement threshold $r^{(\odot)}$ on the mixed collection $B_{n-1} \cup B_{n-2}$ (Corollary 13, formerly Proposition 13). The general theorem additionally fires on non-singleton partition pairs at $k < n - 1$, where the three specializations are vacuous. A structural obstruction (Corollary 19): a bounded peak queue $k < n - 1$ simultaneously blocks the three computable specializations but not the partition-pair mechanism via non-singleton splits. Together the four analytic thresholds certify 66 of the 67 observed empty Cores in our 175-instance grid (the remaining one near-complement case eludes the closed-form thresholds by $\approx 2 \times 10^{-3}$), with no false positives across 525 policy-instance pairs (three dispatch policies). An asymptotic refinement (Theorem 16) ties $r^{**} \rightarrow 1$ to the classical Beardwood–Halton–Hammersley $\sqrt{n}$ scaling of the Euclidean TSP, and a scale-up to $n = 50$ confirms the asymptotic decay.

Scope and methodological choice

The paper is scoped to a single-vehicle routing game. This is a deliberate methodological choice — the single-tour setting isolates the temporal coalition-feasibility structure of $\mathcal{F}$ from fleet-assignment confounds, and the complement-based arguments rely on the single-vehicle insertion identity. A multi-vehicle *Online Vehicle Routing Game* extending this framework with capacity constraints is identified as future work in Section 7.

The manuscript has not been published elsewhere and is not under consideration at any other journal.

Reproducibility

All code, random seeds, and raw CSV results are at <https://github.com/sage-Tae/online-tsg-2026>. The repository ships version tags for each iteration in the revision history. `REPRODUCIBILITY.md` documents the environment, the full 175-instance main study (Table 4 of the manuscript), the 45-instance scale-up to $n = 50$, and the partition-pair-dual analysis script `code/scripts/intermediate_dual_analysis.py` used to extract the 9 partition-pair certificates of Supplementary §S9.

Declarations

- Declaration of interest: The authors declare no competing financial interests or personal relationships that could have influenced this work.
- The work reported has not been published elsewhere, and is not under consideration at any other journal.
- All authors have read and approved this submission.
- We disclose that Claude (Anthropic) was used to assist with language editing and analysis during manuscript preparation, as declared in the manuscript.

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We thank the editorial office in advance for handling this manuscript and look forward to the reviewers' feedback.

Sincerely,

Hyunchul Tae, on behalf of the authors Korea Institute of Industrial Technology
sage@kitech.re.kr ORCID: 0000-0002-2277-0722